BugForge: Constructing and Utilizing DBMS Bug Repository to Enhance DBMS Testing

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Introduction

A bug report serves as the basic entry or record within a bug repository, containing essential information about a specific bug.

- include a natural language description of the problem, version, environment information, logs, and typically an attached PoC.

A PoC (Proof of Concept) is a minimal artifact that demonstrates a bug or vulnerability.

- sequences of SQL statements that expose crashes, inconsistencies, or incorrect query results.

Basic Data:
#ID: 110804 **#Summary:** sql can't be logged in the slow query log.
#Status: Verified **#Severity:** S3 **#Component:** Logging #...

Whole Description:
When the variable min_examined_row_limit is set as a number larger than 0...
How to Repeat: **Set min_examined_row_limit larger than 0 such as 100.**
Execute a query as **"select count(*) from table_name"** which take longer time than long_query_time. The sql won't be logged in the slow query log.

Comments:
Reproduced as described. The rows_examined for the query “select count(*) from table_name” should not be calculated.

Fig. 1: MySQL Bug Report #110804

```
DELIMITER ^ ...
CREATE FUNCTION bug.test() RETURNS text
BEGIN
   SIGNAL SQLSTATE '45000';
   RETURN "returned from bug.test function";
END;^ ...
```

Fig. 2: PoC in MySQL bug report #102205

Motivation

Bug repositories are crucial in industry practice.

- They store bug-triggering information, especially PoCs, that reproduce failures.

Bug repositories are incomplete, unstructured, and hard to execute automatically.

- Incomplete and ambiguous bug reports;
- Limited support for structured analysis;
- Siloed bug information across DBMSs.

This paper proposes *BugForge*, a framework that constructs standardized DBMS bug repositories and leverages them to generate high-quality test cases to enhance DBMS testing.

Challenge

C1 (难抽取): Mining bug information from **unstructured** and **evolving** DBMS bug reports that interleave SQL semantics with naturallanguage descriptions.

C2 (难重建): Restoring **executability** and **preserving semantic richness** to construct high-quality test cases that depend on specific database states.

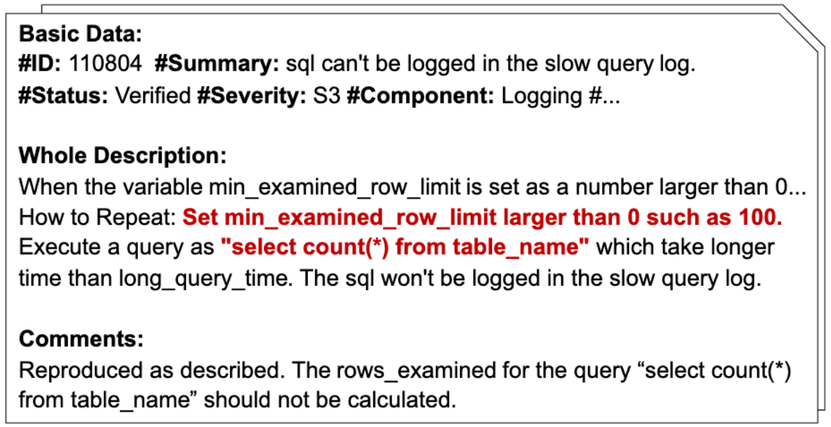

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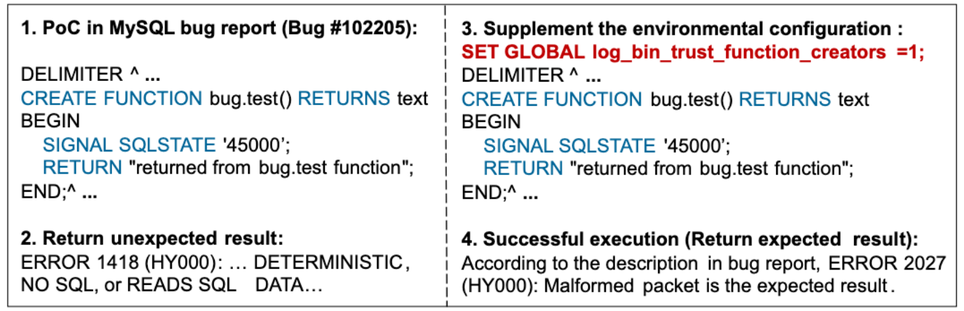
Fig. 1: MySQL Bug Report `#110804` contains a raw PoC that is difficult to extract automatically. This report complicates automated analysis, as the PoC is conveyed in prose instead of being explicitly presented in a distinct code block. The sequence of statements to reproduce the bug, must be manually reconstructed from the content of bug report.



1. PoC in MySQL bug report (Bug #102205):

```
DELIMITER ^ ...
CREATE FUNCTION bug.test() RETURNS text
BEGIN
  SIGNAL SQLSTATE '45000';
  RETURN "returned from bug.test function";
END;^ ...
```

2. Return unexpected result:
ERROR 1418 (HY000): … DETERMINISTIC, NO SQL, or READS SQL DATA…

3. Supplement the environmental configuration :

```
SET GLOBAL log_bin_trust_function_creators =1;
DELIMITER ^ ...
CREATE FUNCTION bug.test() RETURNS text
BEGIN
  SIGNAL SQLSTATE '45000';
  RETURN "returned from bug.test function";
END;^ ...
```

4. Successful execution (Return expected result):
According to the description in bug report, ERROR 2027 (HY000): Malformed packet is the expected result.

Fig. 2: MySQL Bug Report `#102205` misses the configuration required for successful execution. The bug is unreproducible because the bug report lacks the configuration information for the `log_bin_trust_function_creators` system variable, but it triggers the expected result after *BugForge* adapts the raw PoC.

Overview

The basic idea of BugForge is to use syntax-aware processing to extract raw PoCs and transform them into high-quality test cases through semantic-guided adaptation.

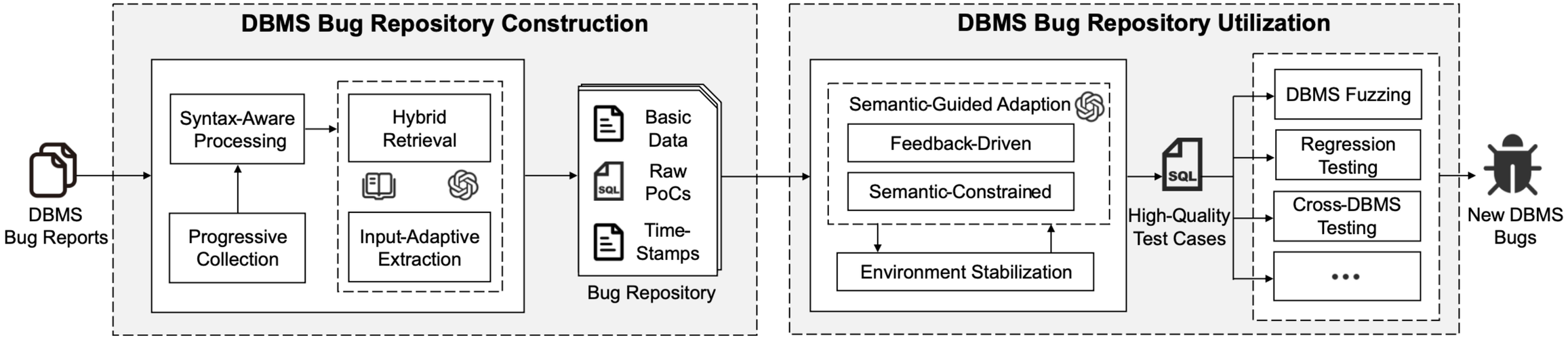


Fig. 3: **The overview design of *BugForge*.** *BugForge* is an end-to-end framework for DBMS bug repository construction and utilization. It progressively collects bug reports, applies syntax-aware processing and the input-adaptive LLM pipeline with RAG to extract raw PoCs and constructs a DBMS bug repository. Next, *BugForge* converts raw PoCs into high-quality test cases through semantic-guided adaptation under a stable environment and subsequently employs these cases in DBMS testing (e.g., fuzzing, regression, and cross-DBMS testing).

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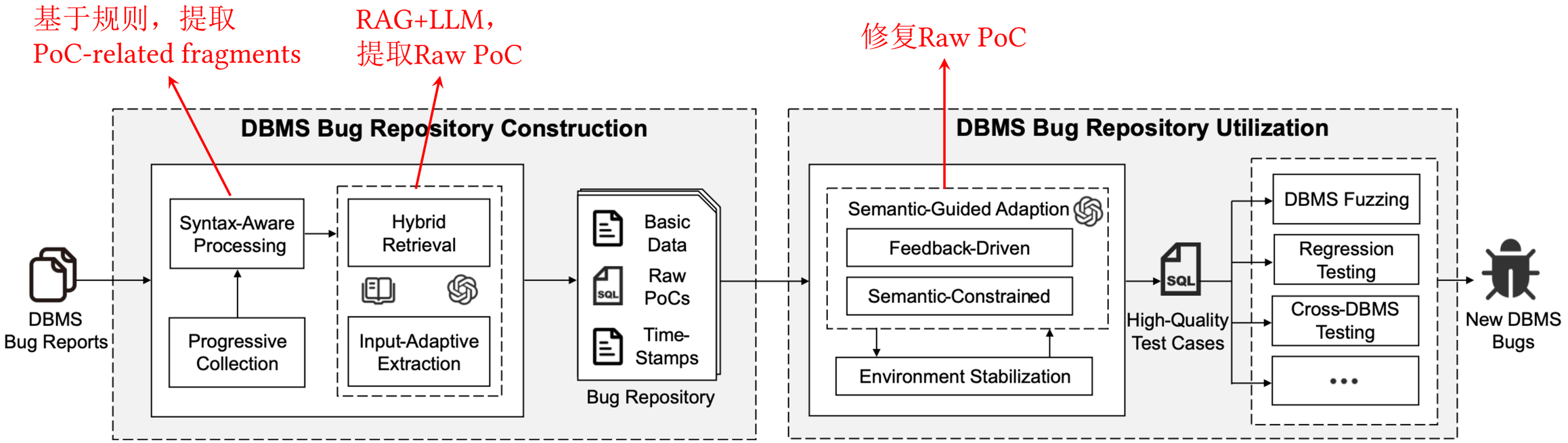

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Syntax-Aware DBMS Bug Report Processing

PoC-related fragments: report fragments that may preserve procedural or semantic clues for constructing a raw PoC, but do not necessarily constitute a complete or directly high-quality raw PoC.

Algorithm 1: Syntax-Aware Bug Reports Processing

Input: DBMS bug report: $\mathcal{L}$.
Output: PoC-related fragments: $\mathcal{P}$, or **null**.

$\mathcal{S} \leftarrow$ an empty list of (index, content) tuples;
$i \leftarrow 0$;
while $i < |\mathcal{L}|$ **do**
  // Capturing Formatted SQL via Text Structure
  if $\mathcal{L}[i]$ *is a start-of-block tag (e.g., markdown)* **then**
    $j \leftarrow$ Find corresponding end tag from $i+1$;
    if j *is found* **then**
      Add $(i, \mathcal{L}[i+1:j])$ to $\mathcal{S}$; $i \leftarrow j$;
  // Capturing Fragmented SQL via SQL Feature
  else if $\mathcal{L}[i]$ *is a endpoint (e.g., semicolon)* **then**
    $j \leftarrow i$; $b \leftarrow 0$;
    while $j \geq 0$ **do**
      $b \leftarrow b + \text{Count}(\mathcal{L}[j],')') - \text{Count}(\mathcal{L}[j],'(')$;
      if *IsSQLSubject*$(\mathcal{L}[j])$ ***and*** $b \leq 0$ **then break**;
      $j \leftarrow j - 1$;
    Add $(j, \mathcal{L}[j:i+1])$ to $\mathcal{S}$;
  // Capturing Implicit SQL via Weight Calculation
  else if *CalculateScore*$(\mathcal{L}[i]) \geq \theta_{score}$ **then**
    Add $(i, [\mathcal{L}[i]])$ to $\mathcal{S}$;
  $i \leftarrow i + 1$;
return $\mathcal{P}$ if $\mathcal{P}$ is not **null** else **null**;

Given a DBMS bug report *L*, *BugForge* scans *L* using a prioritized **three-stage strategy** that progressively captures SQL code representations ranging from explicitly formatted blocks to fragmented or implicit statements.

阶段一：通过文本结构捕获格式化的 SQL
Markdown fenced block: ``` ... ```
Jira code block: {code:xxx} ... {code}

阶段二：通过SQL特性捕获分片化SQL
基于分号结尾的逆推
括号平衡约束等

阶段三：通过权重计算捕捉隐式 SQL

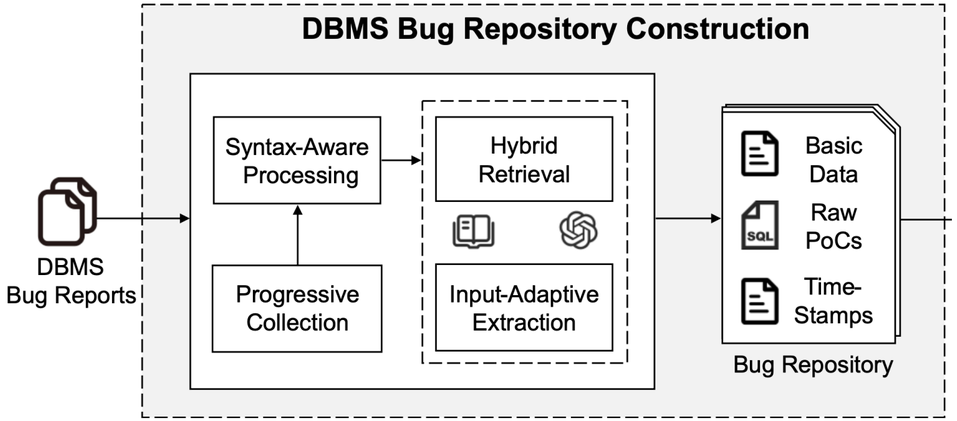
加分项
以分号结尾：+7
行首命中 SQL 主语：+18
命中方言关键字：每次 +3，最多 +14
...

减分项
绝对噪声，如 URL、十六进制地址
各种结构化噪声：Query OK、Records:
自然语言特征越多，扣分越多
...

Input-Adaptive Raw PoC Extraction

PoC-related fragments are incomplete, or semantically ambiguous.

To improve extraction accuracy while controlling context overhead, *BugForge* employs an LLM-based extraction framework with hybrid retrieval and adaptive input expansion.



(1) System Prompt: You are a PoC analysis and SQL extraction tool in DBMS.

(2) Extraction Prompt: You must follow this Three -Step Thinking process:
Step 1: Context-Guided Extraction: Extract the raw PoC from input content.
Step 2: On-Demand Expansion: Request extra bug report content only when needed.
Step 3: Self-Examination: Recheck whether the extraction follows these rules:

#R1: Preserve bug-triggering semantics. #R2: Remove irrelevant content and noise.
#R3: Do not invent unsupported details. #R4: Output N/A if the raw PoC is unextractable.

(3) Reference Prompt: (Hybrid-retrieved references from a dynamically evolving library)
Positive: (#Bug Summary #PoC-related Fragments) -> Chain-of-Thought -> raw PoC
Negative: (#Bug Summary #PoC-related Fragments) -> Chain-of-Thought -> N/A

(4) Task Prompt: Now, analyzing the current report with:
#Summary #PoC-related Fragments #Retrieved References. Request more content if needed.

Fig. 4: **An example prompt for LLM in raw PoC extraction.** It consists of four components: (1) System Prompt, which instructs LLM to analyze bug reports. (2) Extraction Prompt, which guides the LLM through extraction, on-demand context expansion, and self-examination. (3) Reference Prompt, which retrieves positive and negative exemplars from the library. (4) Task Prompt, which supplies the bug summary, raw PoC-related fragments, and retrieved references for the target report.

从动态维护的 reference library 中检索相似 exemplar。
检索方式是 dense retrieval (保证语义相似) + 关键词检索 (强调 DBMS 特有的词法特征，比如 SQL operator、error message)。

Input-Adaptive Raw PoC Extraction

Finally, after extracting a raw PoC, *BugForge* evaluates its quality in terms of **contextual consistency** and **structural completeness**.

- High-confidence cases are then distilled into reusable exemplars and inserted back into the RAG library.
- The validated raw PoC is then integrated with report metadata to construct the DBMS bug repository.

语义一致性得分方式：

1. 从参考 SQL 和候选 SQL 中抽取一组 semantic anchors，例如 objects/qualified_columns/functions/features/data_types等。
2. 对每个anchor分别计算： ***coverage*** = |ref ∩ cand| / |ref|,
   jaccard = |ref ∩ cand| / |ref ∪ cand|。
   每个anchor的得分: ***anchor_score*** = 0.7 * coverage + 0.3 * jaccard
3. 全局得分=加权平均所有anchor_score

#ID: BUG #18840
#Report Link: https://www.postgresql.org/message -id/... **#...**
#Summary: Segmentation fault in executing select unnest(array(oidvector)).
#Description: The total content of this bug... **#Comment：**The responses...
#CollectedDate: YYYY-MM-DD, reference timestamp for subsequent updates.
#Bug Type: (Crash√, Performance ×, Logic×, Unkown×)
#PoC-related Fragments: Report Fragments for constructing the raw PoC.

#Raw PoC:

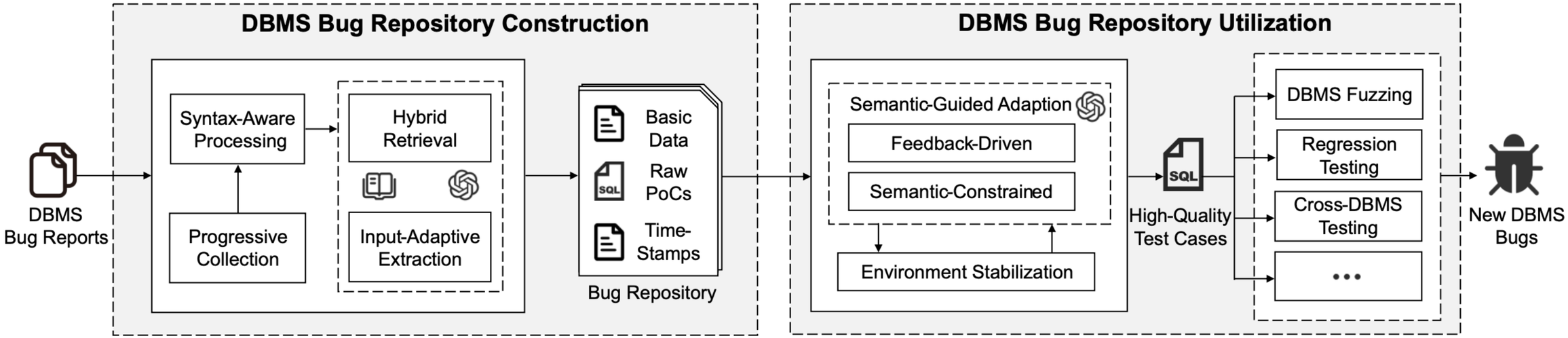
```
CREATE TABLE my_table ( id SERIAL PRIMARY KEY, oidvector_col oidvector );
INSERT INTO my_table ( oidvector_col ) VALUES ( '12345 67890 54321' );
SELECT unnest ( ARRAY( SELECT oidvector_col )) from my_table;
```

Fig. 5: **An example case in *BugForge* repository.** *BugForge* normalizes heterogeneous DBMS bug reports into a structured representation. Each case includes: (1) basic metadata of the report, (2) summarized bug-type labels and the PoC-related fragments, (3) the extracted raw PoC used for downstream adaptation and DBMS testing.

Bug Repository Utilization

Object: convert raw PoCs into high-quality test cases and then leverages them for automated DBMS testing.

High-quality test cases: executability + semantic richness.



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Semantic-Guided Test Case Adaptation

1. **Feedback-driven adaptation:** diagnoses execution-blocking issues through runtime feedback and repairs these.

根据人为构造的轻量级
知识库进行错误类型分类

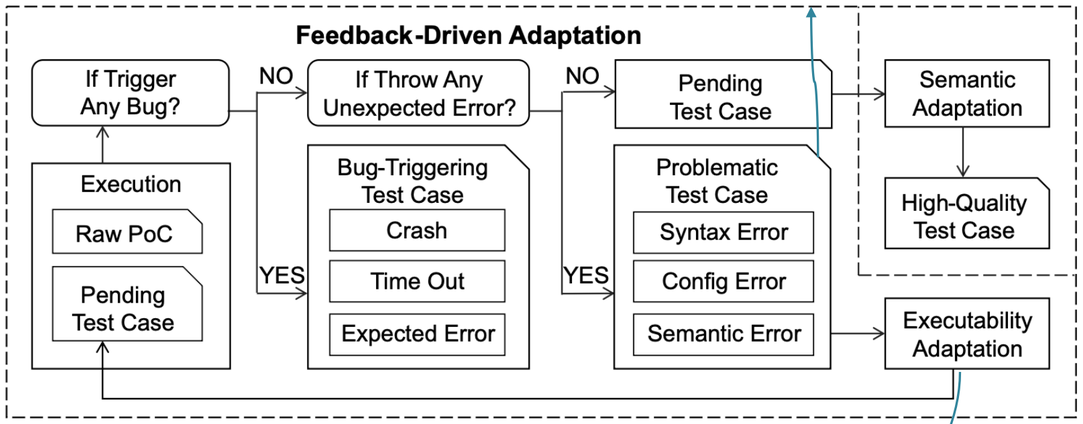

Fig. 6: **Process of Feedback-Driven Adaptation.** Each raw PoC is first executed and classified by its observed behavior. Cases with unexpected errors are further diagnosed using execution feedback and DBMS knowledge, and then repaired through LLM-guided executability adaptation.

```python
# --- 分类规则定义区 (保持不变) ---
CLASSIFICATION_RULES = {
    "Syntax Error": [
        'parse_error', 'syntax_error', 'wrong_name_for', 'dup_fieldname',
        'wrong_paramcount', 'illegal_grant', 'wrong_value_count', 'bad_sqlstate',
        'wrong_arguments', 'must_have_alias', 'unacceptable mutual recursion',
        'window specification', 'with column list', 'syntax', 'incorrect usage',
        'incorrect parameter', 'incorrect name', 'incorrect definition',
        'malformed'
    ],
    "Semantic Error": [
        'dup_entry', 'dup_key', 'no_such_table', 'bad_table_error', 'bad_field_error',
        'table_exists_error', 'no_referenced_row', 'row_is_referenced',
        'constraint_failed', 'unknown table', 'unknown column', 'table exists',
        'database exists', 'column count doesn', 'subquery returns more than 1 row',
        'ambiguous', 'unknown database', 'key column', 'doesn\'t exist',
        'not found', 'is not', 'already exists'
    ],
```

把 diagnosis 的结果（错误类型）+ runtime feedback（报错信息）
一起提供给 LLM，引导它补充缺失信息并修复执行阻塞问题

Semantic-Guided Test Case Adaptation

Semantic-constrained adaptation: uses semantic anchors captured from raw PoCs to guide adaptation towards results that do not lose the semantic richness of raw PoCs.

1. **Anchor match** is used to constrain the LLM to keep the core data dependencies of the original raw PoCs.
2. **Key coverage** aims to restrict LLM from modifying key SQL operations.
3. **Rewrite bound** is designed to prevent the LLM from excessively rewriting the original query structure.

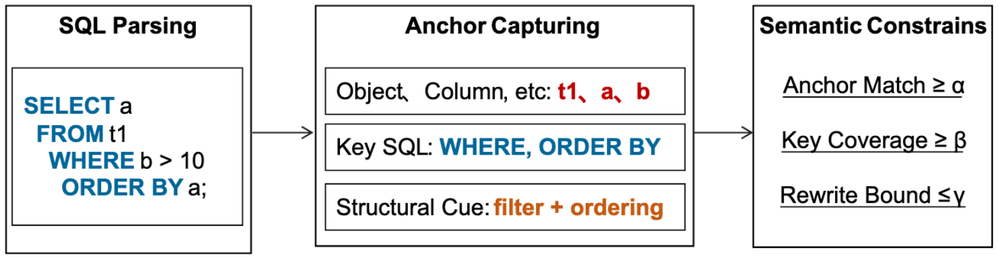

Fig. 7: **Process of Semantic-Constrained Adaptation.** *BugForge* first extracts semantic anchors from raw PoC, including data dependencies, SQL keywords, and structural cues, and organizes them into an anchor profile. Based on it, *BugForge* applies three constraints, namely anchor match, key coverage, and rewrite bound, to guide the LLM in preserving as much semantic richness in raw PoCs as possible during adaptation.

Rewrite Bound:

1. 把参考 SQL 和候选 SQL 做 token 化，计算它们的**序列相似度**：

   seq_similarity = SequenceMatcher(ref_tokens, cand_tokens).ratio()

2. 定义**改写强度**：

   rewrite_bound = 1 - seq_similarity

Environment Stabilization

To strictly guarantee execution independence while maximizing throughput, *BugForge* employs a statement-driven hierarchical impact assessment mechanism.

1. Parse input into atomic SQL statements.
2. Extract key semantics (verb, object, scope).
3. Map statements to risk levels：

   ① NONE（普通语句，对环境影响有限，不做额外补偿）

   ② RESTART（会改变运行时状态，但通常不需要彻底重建环境）

   ③ RESET（会污染全局环境、权限、插件、用户、数据库元信息，执行后需要强隔离恢复）

以 MariaDB 为例，见 mariadb/dry_run.py (line 560)：

- 如果出现这些语句，直接返回 `RESET`
  - `CREATE USER|ROLE`
  - `DROP USER|ROLE`
  - `ALTER USER|ROLE`
  - `GRANT`
  - `REVOKE`
  - `INSTALL PLUGIN|SONAME`
  - `UNINSTALL PLUGIN|SONAME`
  - `FLUSH PRIVILEGES`
- 如果没有命中上述 full-reset 语句，但命中：
  - `SET GLOBAL`
  - `SET PERSIST`

  那么返回 `RESTART`
- 否则返回 `NONE`

Evaluation

- RQ1（repository 的质量）: What are the characteristics and utilizability of the bug repository constructed by BugForge?
- RQ2（发现新 bug 的能力）: Can BugForge help discover new DBMS bugs?
- RQ3（与现有方法的对比）: How does BugForge compare with the state-of-theart DBMS testing tools?
- RQ4（test case 本身的价值）: How do high-quality test cases in BugForge compare with others in terms of DBMS testing?

Tested DBMSs: MySQL, PostgreSQL, MariaDB, MonetDB.

LLM-assisted stages: Gemini-2.5-flash for raw PoC extraction and Gemini-2.5-pro for the more complex adaptation tasks.

RQ1: Assessment of *BugForge* Bug Repository

Characteristic of Bug Repository.

- Collect all publicly available bug reports spanning from 1998 to 2026.

Utilizability of Bug Repository.

- Assess the usability by evaluating the coverage of SQL features and the number of adapted PoCs it contains.

TABLE I: Statistics of Bug Repository Built by *BugForge*

	MySQL	**MariaDB**	**PostgreSQL**	**MonetDB**	**Total**
Covered Time	2003-2026	2009-2026	1998-2026	2011-2026	–
Collected Reports	18,212	12,442	4,163	2,815	37,632
Extracted Raw PoCs	17,723	11,581	3,758	2,468	35,530
CVEs	2,055	147	364	31	2,597
high-quality Test Cases	16,112	11,426	3,616	2,444	33,598

TABLE II: Covered Data Types and Keywords in DBMSs

DBMS	**Data Type**			**Keyword**		
	Total	Covered	Rate	Total	Covered	Rate
MySQL	33	33	100.00%	754	666	88.33%
MariaDB	36	35	97.22%	683	616	90.19%
MonetDB	30	28	93.33%	301	268	89.04%
PostgreSQL	68	52	76.47%	491	452	92.06%
Total	**167**	**148**	**88.62%**	**2229**	**2002**	**89.82%**

RQ1: Assessment of *BugForge* Bug Repository

High-quality Test Cases Generated by BugForge.

- High-quality test cases refer to problematic raw PoCs that successfully pass both feedback-driven adaptation and semantic-constrained adaptation, meaning that they are executable and satisfy the semantic anchor constraints.

TABLE III: Number of High-quality Test Cases Adapted from Problematic Test Cases.

	MySQL	**MariaDB**	**PostgreSQL**	**MonetDB**	**Total**
Problematic Cases	12,504	4,523	1,911	292	**19,230**
Adaptable Cases	10,893	4,368	1,769	268	**17,298**
Ratio	87.12%	96.57%	92.57%	91.78%	**89.95%**

RQ2: Bugs Detected With *BugForge*

Three types of testing: DBMS fuzzing, DBMS regression testing, and cross-DBMS verification.

For *BugForge*, we use the configuration where *Griffin* is seeded with both built-in unit test cases and the high-quality test cases generated from *BugForge*.

TABLE IV: List of previously unknown bugs detected by DBMS testing with *BugForge* employed

#	DBMS	ID	Status	Component	Testing Method	Description
1	MySQL	S2310126	Confirmed	Function	DBMS Fuzzing	MySQL 9.4.0 crashes at Item_func_bit::val_int.
2	MySQL	S2310135	Confirmed	Function	DBMS Fuzzing	MySQL 9.4.0 crashes at Item_func::print_op.
3	MySQL	S2310142	Confirmed	Function	DBMS Fuzzing	MySQL 9.4.0 crashes at Item_func::walk.
4	MySQL	S2310157	Confirmed	Item	DBMS Fuzzing	MySQL 9.4.0 crashes Item_ref::walk.
5	MySQL	S2310161	Confirmed	Item	DBMS Fuzzing	MySQL 9.4.0 crashes at Item_ref::val_int.
6	MySQL	S2310174	Confirmed	Filed	DBMS Fuzzing	MySQL 9.4.0 crashes at Field::set_notnull.
7	MySQL	S2310188	Confirmed	Field	DBMS Fuzzing	MySQL 9.4.0 crashes at set_field_to_null_with_conversions.
8	MySQL	S2310190	Confirmed	Optimizer	DBMS Fuzzing	MySQL 9.4.0 crashes at Query_expression::optimize.
9	MySQL	S2310208	Confirmed	View	DBMS Fuzzing	Server crashes at Item_view_ref::check_column_privileges.
10	MySQL	# 119085	Reported	Server	Cross-DBMS Testing	FLUSH PRIVILEGES crash after table column drop.
11	MySQL	# 119086	Reported	Stored Routines	Cross-DBMS Testing	Server crash after VIEW-to-TABLE dependency change.
12	MariaDB	MDEV-37625	Confirmed	Server	DBMS Fuzzing	MariaDB 12.1.1-rc crashes at alloc_root.
13	MariaDB	MDEV-37628	Confirmed	Optimizer	DBMS Fuzzing	MariaDB 12.1.1-rc crashes at find_field_in_tables.
14	MariaDB	MDEV-37640	Confirmed	JSON/Server	DBMS Fuzzing	MariaDB 12.1.1-rc crashes at String::append.
15	MariaDB	MDEV-37641	Confirmed	Optimizer	DBMS Fuzzing	MariaDB 12.1.1-rc crashes at sub_select_postjoin_aggr.
16	MariaDB	MDEV-37646	Confirmed	Optimizer	DBMS Fuzzing	MariaDB 12.1.1-rc crashes at Item::save_decimal_in_field.
17	MariaDB	MDEV-37647	Confirmed	JSON/Optimizer/Server	DBMS Fuzzing	Server crashes at set_field_to_null_with_conversions.
18	MariaDB	MDEV-37648	Confirmed	Optimizer/Server	DBMS Fuzzing	Server crashes at Item_direct_view_ref::val_decimal.
19	MariaDB	MDEV-37768	Reported	Server	Cross-DBMS Testing	Server crash when mysql.procs_priv has an invalid structure.
20	MariaDB	MDEV-37671	Confirmed	Optimizer/Server	Regression Testing	Solved bug MDEV-32326 reappeared in 12.1.1-rc.
21	MariaDB	MDEV-37735	Confirmed	Sequences	Regression Testing	Solved bug MDEV-37172 reappeared in 11.8.4 and 12.1.1-rc.
22	MariaDB	MDEV-37734	Reported	Optimizer	Regression Testing	Solved bug MDEV-32308 reappeared in 11.8.4 and 12.1.1-rc.
23	MonetDB	# 7720	Confirmed	Server	DBMS Fuzzing	MonetDB Mar2025-SP2 server crashes at stmt_cond.
24	MonetDB	# 7721	Reported	Server	DBMS Fuzzing	MonetDB Mar2025-SP2 server crashes at stmt_replace.
25	MonetDB	# 7722	Fixed	Server	DBMS Fuzzing	MonetDB Mar2025-SP2 server crashes at rel_with_query.
26	MonetDB	# 7723	Reported	Server	DBMS Fuzzing	MonetDB Mar2025-SP2 server crashes at rel_has_freevar.
27	MonetDB	# 7724	Reported	Server	Regression Testing	Solved bug # 3009 reappeared in Mar2025-SP2 release.
28	MonetDB	# 7725	Fixed	Server	Regression Testing	Solved bug # 7482 reappeared in Mar2025-SP2 release.
29	MonetDB	# 7726	Reported	Server	Regression Testing	Solved bug # 7486 reappeared in Mar2025-SP2 release.
30	MonetDB	# 7727	Fixed	Server	Regression Testing	Solved bug # 7436 reappeared in Mar2025-SP2 release.
31	MonetDB	# 7728	Reported	Server	Regression Testing	Solved bug # 7488 reappeared in Mar2025-SP2 release.
32	MonetDB	# 7729	Confirmed	Server	Regression Testing	Solved bug # 7472 reappeared in Mar2025-SP2 release.
33	PostgreSQL	# 19055	Confirmed	Parser	DBMS Fuzzing	The count() aggregate is assigned the wrong agglevelsup.
34	PostgreSQL	# 19382	Fixed	Executor	DBMS Fuzzing	PostgreSQL 14-17.7 server crash at __nss_database_lookup.
35	PostgreSQL	# 19384	Reported	Parser	DBMS Fuzzing	PostgreSQL 17.7 server crashes at textout.

RQ3: Comparison of SOTA DBMS Testing Tools

The high-quality test cases generated from DBMS bug reports can guide testing toward deeper and broader execution paths than purely generator-based approaches.

TABLE V: Comparison of Typical DBMS Testing Tools.

DBMS	Covered Branches			Detected Bugs		
	SQLancer	SQLsmith	BugForge	SQLancer	SQLsmith	BugForge
MySQL	106,971	96,311	209,720	5	3	8
MariaDB	46,298	30,896	140,677	4	2	7
PostgreSQL	48,030	50,285	90,763	2	1	3
MonetDB	N/A	33,209	83,420	N/A	2	5
Total†	201,299	177,492	441,160	11	6	18

RQ4: Test Case Quality

U: built-in unit test cases from DBMS repositories.

B: *BugForge.*

Notably, the *BugForge* corpus contains only about 14.2 MB of total file content, compared with 100.3 MB for the unit-test corpus, corresponding to an 85.9% reduction.

TABLE VI: Covered Branches with different initial seeds.

DBMS	GRIFFIN^{U}	GRIFFIN^{B}	GRIFFIN^{U+B}	SQUIRREL^{U}	SQUIRREL^{B}	SQUIRREL^{U+B}
MySQL	182,709	192,710	209,720	147,936	159,794	175,794
MariaDB	114,029	134,595	140,677	89,115	100,188	109,318
PostgreSQL	73,273	85,862	90,763	59,253	63,235	74,828
MonetDB	32,990	79,797	83,420	N/A	N/A	N/A
Total	403,001	492,964	524,580	296,304	323,217	359,940
Increment	–	+22.32%	+30.17%	–	+9.08%	+21.48%

TABLE VII: Detected Bugs with different initial seeds.

DBMS	GRIFFIN^{U}	GRIFFIN^{B}	GRIFFIN^{U+B}	SQUIRREL^{U}	SQUIRREL^{B}	SQUIRREL^{U+B}
MySQL	4	6	8	3	5	6
MariaDB	3	5	7	2	4	5
PostgreSQL	1	2	3	1	1	2
MonetDB	2	5	5	N/A	N/A	N/A
Total	10	18	23	6	10	13
Increment	–	+80.00%	+130.00%	–	+66.67%	+116.67%

Conclusion

Problem: Bug reports contain valuable PoCs but are not directly usable.

Approach: BugForge builds and utilizes a unified bug repository.

Key Idea: Transform raw PoCs into executable, high-quality test cases.

Result: Improved coverage and discovered new bugs.